

7. Dual-Mode Operation (User Mode vs Kernel Mode)

Computer Science Fundamentals | Module 2: OS Protection | CS_Fundamental-00

Reference to Previous Concepts: Concept 3.1 & 3.2 mein User View vs System View dekha tha, aur Concept 4.1 mein Kernel ko bridge dekha tha. Yeh topic **Image 7 (bottom) & Image 8 (top)** par aadharit hai jo **User Mode vs Kernel Mode** aur code `printf("Hello");` ka transition explain karta hai.

7.1 Dual-Mode Operation ki Zaroorat (Protection & Safety)

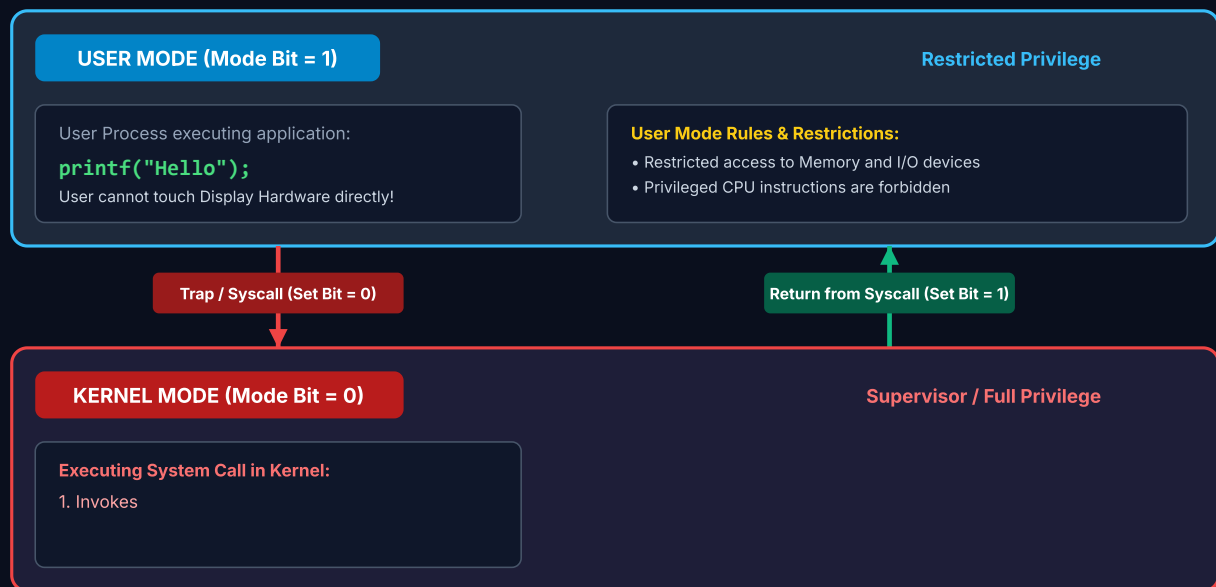
Agar koi normal software hardware ya RAM ko directly modify kar sake, toh ek glitch poore computer ko crash kar sakta hai. Isliye hardware do distinct privilege modes provide karta hai.

7.2 User Mode vs Kernel Mode

- **User Mode (Restricted Privilege - Mode Bit = 1):** User processes execute hote hain. Hardware resources direct access nahi kiye ja sakte.
- **Kernel Mode (Supervisor / Full Privilege - Mode Bit = 0):** OS Kernel execute hota hai. Hardware, RAM, registers aur privileged instructions ka full access hota hai.

Fig 7.1: Dual-Mode Operation & System Call Transition

Reference: Image 7 (Bottom) & Image 8 (Top) [User Mode → Kernel Mode via printf]



`sys_write()` 2. Sends data to Video Buffer / Display 3. Writes "Hello" to Terminal Screen Kernel Mode Privileges: • Full control over hardware resources & memory • Direct execution of privileged machine instructions • Manages hardware interrupts, I/O ports & MMU

Fig 7.1: Dual Mode Transition & System Call Trace

7.3 Handwritten Example & Step-by-Step Thought Process: printf("Hello");

Image 7 bottom note: USER MODE -----> Kernel mode | printf("Hello");

Thought Process Trace:

1. User Mode Execution (Mode Bit = 1):

User program `printf("Hello")` call karta hai. Library pehchanti hai ki display screen par write karna ek hardware operation hai jo User Mode mein allowed nahi hai.

2. Trap & Mode Switch (1 → 0):

Program `sys_write()` system call execute karta hai. Hardware Trap trigger hota hai aur CPU Mode Bit ko `1` se `0` (Kernel Mode) switch kar deta hai.

3. Kernel Mode Execution (Mode Bit = 0):

Kernel parameters verify karta hai aur display controller ke video buffer mein safely write karta hai. Screen par "Hello" print ho jata hai.

4. Return to User Mode (0 → 1):

Kernel return instruction deta hai. Hardware Mode Bit ko wapas `1` (User Mode) set kar deta hai aur user app aage chalta hai.

7.4 User Mode vs Kernel Mode Comparison Table

Paimana	User Mode	Kernel Mode
Mode Bit	1	0
Privilege	Restricted Access	Full Hardware & Memory Access
Executing Code	User Applications (Chrome, Word, Python)	Operating System Kernel
Hardware I/O	Direct I/O Forbidden (Needs System Call)	Direct I/O Permitted